VEERMATA JIJABAI TECHNOLOGICAL INSTITUTE

Department of Computer Engineering

OPTIMIZATION TECHNIQUES LABORATORY - QUESTION 4

Cold-Chain Delivery Job Assignment Using ML, QAOA and Parallel Computing

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GitHub Repository: https://github.com/sofiaabidi/ot_q4_241071072

1. Aim

To assign six delivery jobs (J1-J6) to two refrigerated vehicles such that the difference between total predicted delivery times is minimized. Every job must be assigned exactly once, and each vehicle must receive 2 to 4 jobs.

2. Problem Formulation

Let $x_i = 1$ if job i is assigned to Vehicle 1, otherwise 0. If t_i is the predicted delivery time, then **T1 = sum($t_i x_i$)** and **T2 = sum($t_i(1 - x_i)$)**. The objective is to **minimize |T1 - T2|**, subject to **2 <= sum(x_i) <= 4**.

3. Methodology

Machine Learning: Generated 100 records using the prescribed seed (72) and features: distance, load and traffic. Trained linear regression, evaluated on a held-out set and predicted times for J1-J6.

Quantum Computing: Formulated the balancing objective as a QUBO/Ising model and used a six-qubit, two-layer QAOA statevector simulation with 4,096 seeded measurements and feasibility filtering.

Parallel Computing: Evaluated all 64 binary assignments using two worker processes; exactly 50 satisfy the job-count constraint. Compared the result with serial exhaustive search and a longest-job-first greedy baseline.

4. Results and Observations

Regression accuracy: MAE = **0.048516 h**; R^2 = **0.992067**. Predicted delivery times (hours):

J1	J2	J3	J4	J5	J6
2.7916	4.6941	3.3732	4.9546	3.8716	4.2437

Method	Vehicle 1	Vehicle 2	T1 (h)	T2 (h)	|T1 - T2| (h)
Greedy	J3, J4, J5	J1, J2, J6	12.1994	11.7294	0.470014
QAOA	J2, J3, J5	J1, J4, J6	11.9389	11.9899	0.050972
Parallel exact	J2, J3, J5	J1, J4, J6	11.9389	11.9899	0.050972

Observation: QAOA matched the exact minimum imbalance of **0.050972 h** (about 3.06 minutes), compared with 0.470014 h for greedy. Both tied optimal masks (22 and 41) were recorded; mask 22 was observed 168 times in QAOA sampling.

5. Verification

All **9 pytest tests passed**. QUBO/Ising equivalence was checked for all 64 assignments, statevector normalization was verified, both parallel worker processes executed, and serial and parallel exhaustive results agreed. A clean-clone run also passed.

6. Conclusion

The implementation successfully combines prediction, quantum optimization and parallel search for the given job-assignment problem. The sampled QAOA assignment achieved the same optimal balance as exact search. The experiment uses generated training data and an ideal statevector simulator; runtimes depend on the local computer.

Evidence: results/runs/20261106T054359Z-f35d/ **To run:** pip install -r requirements.txt; python main.py; python -m pytest -q